This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Temporal Interaction in Spiking Transformers with Multi-Delay Mixer

Kexin Shi[1] Hanwen Liu[1] Zeyang Song[2] Yang Liu[1] Jieyuan Zhang[1]
Shuai Wang[1] Jibin Wu[3] Malu Zhang[1,*] Yang Yang[1]
[1]University of Electronic Science and Technology of China
[2]National University of Singapore
[3]Hong Kong Polytechnic University
[*]Corresponding author: maluzhang@uestc.edu.cn

Abstract

*Spiking Neural Networks (SNNs) have gained significant attention due to their event-driven computational paradigm, making them promising for neuromorphic computing. In recent years, the integration of SNNs and Transformer architectures has made remarkable progress in various tasks. However, existing spiking self-attention mechanisms predominantly focus on spatial information while neglecting explicit temporal modeling, leading to suboptimal performance. In this paper, we introduce the Temporal Interaction Coefficient (TIC) to analyze temporal dependency patterns in these spatial-only attention mechanisms, revealing their limited temporal interactions and restricted pattern diversity. To overcome this issue, we propose the **M**ulti-**D**elay **Mixer** (**MD-Mixer**), drawing inspiration from time delay mechanisms in the nervous system. Specifically, MD-Mixer introduces multiple temporal delays to perform effective time mixing and facilitate temporally enriched spatial attention. In addition, it can be integrated seamlessly into existing Spiking Transformers as a drop-in replacement while maintaining computational efficiency. Extensive evaluations on static and neuromorphic benchmarks demonstrate that MD-Mixer substantially improves the performance of Spiking Transformers, outperforming existing state-of-the-art (SOTA) methods. This work establishes MD-Mixer as an effective and general solution for temporal modeling in event-driven architectures.*

1. Introduction

Spiking Neural Networks (SNNs) have emerged as a promising paradigm in neuromorphic computing, offering a fundamentally different approach to information processing compared to conventional Artificial Neural Networks (ANNs) [9, 14, 27]. Drawing inspiration from biological neurons, SNNs encode and transmit information through discrete, event-driven spikes, which are triggered when membrane potentials surpass firing thresholds. The sparse and event-driven nature [4] of spike-based computation enables remarkable energy efficiency when deployed on neuromorphic hardware platforms [1, 5, 28, 30]. Moreover, the inherent temporal dynamics [52] embedded in spike timing provide a natural framework for processing spatiotemporal information. These distinctive characteristics establish SNNs as ideal candidates in energy-constrained spatiotemporal computing scenarios [21, 22, 24, 25].

To further unlock their potential, Transformer architectures have recently been integrated into SNNs, spawning numerous Spiking Transformer models [35, 45, 53, 56, 62, 63], demonstrating substantial performance improvements across diverse applications [19, 36, 49, 60]. Despite these advances, existing self-attention mechanisms [11, 38, 41, 51] suffer from a fundamental limitation that they effectively capture spatial relationships within individual timesteps but critically fail to model temporal dependencies. This temporal modeling deficiency stems from the exclusive reliance on intrinsic spiking neuron dynamics [8, 10], which proves insufficient for capturing temporal patterns inherent in neuromorphic data [33, 58]. Existing solutions attempting to address temporal dependencies either incur prohibitive computational costs or sacrifice modeling capacity [18, 34, 48], leaving temporal modeling as a critical unresolved challenge in Spiking Transformers. Compounding this issue, the absence of proper evaluation metrics limits researchers to task-specific performance metrics, impeding both theoretical understanding and practical development of Spiking Transformers.

In this paper, we introduce the *Temporal Interaction Coefficient* (TIC), a quantitative metric that measures temporal modeling capability in attention mechanisms. Through TIC-based analysis, we reveal that existing spiking attention mechanisms exhibit sharply peaked reliance distributions at current timesteps with negligible historical contributions, indicating severely limited temporal interactions. This find-

ing establishes temporal modeling as a critical bottleneck in Spiking Transformers. To address this limitation, we draw inspiration from biological axonal transmission delays and propose the *Multi-Delay Mixer* (MD-Mixer), which introduces multiple learnable delay branches with adaptive channel-wise weighting to explicitly model multi-scale temporal dependencies. We further develop a soft-to-hard optimization strategy with progressive temperature annealing, enabling efficient learning of discrete delays through continuous gradient descent. Building upon MD-Mixer's temporal mixing capability, we construct the Multi-Delay Self-Attention Framework that integrates delay-based temporal interactions with spatial attention. By applying MD-Mixer to compute temporally enriched Keys and Values while maintaining temporally aligned Queries, our framework enables effective space-time mixing that captures both spatial and temporal dependencies in spike-driven computation. Extensive experiments demonstrate that MD-Mixer consistently improves the performance of various Spiking Transformer architectures across static, neuromorphic, and long-sequence benchmarks. Our main contributions are summarized as follows:

- We introduce the *Temporal Interaction Coefficient* (TIC) to quantify temporal modeling capacity in spiking attention mechanisms. TIC analysis reveals that existing mechanisms exhibit sharply peaked reliance distributions at current timesteps with negligible historical contributions, establishing temporal modeling as a fundamental bottleneck in Spiking Transformers.
- We propose the *Multi-Delay Mixer* (MD-Mixer), which employs multiple learnable delay branches and adaptive channel-wise weighting to explicitly capture multi-scale temporal dependencies. We further develop the Multi-Delay Self-Attention Framework that integrates MD-Mixer with spatial attention to achieve effective space-time mixing through delay-based temporal interactions.
- We design a soft-to-hard optimization strategy with progressive temperature annealing that enables efficient learning of discrete delays through continuous gradient descent, facilitating the discovery of optimal temporal delay configurations.
- We conduct comprehensive experiments demonstrating that MD-Mixer consistently improves various Spiking Transformer architectures across static datasets, neuromorphic benchmarks, and long-sequence tasks, establishing new state-of-the-art results while maintaining computational efficiency.

2. Related Work

2.1. Spiking Neural Networks

SNNs fundamentally differ from conventional ANNs by encoding and transmitting information through discrete spikes rather than continuous activations, which endows them with promising advantages in terms of energy efficiency [26, 37] and biological plausibility. The introduction of direct training methods [8, 50] and ANN-to-SNN conversion strategies [2, 12] has enabled the development of high-performance and energy-efficient SNNs [15, 39, 40, 47]. Building on these advances, researchers have incorporated spiking attention mechanisms [55, 59, 64] into SNNs to further enhance recognition performance on ImageNet [6] and neuromorphic datasets [20]. Moreover, the adoption of residual connections [7, 13] has effectively mitigated the gradient vanishing problem in deep SNNs, enabling large-scale network training. Additionally, recent studies have explored integrating Transformer architectures into SNNs, resulting in representative models such as Spikformer [63], SpikingResformer [38], and the Spike-Driven Transformer [53, 57]. These models have further improved the performance of SNNs on image recognition and neuromorphic datasets.

2.2. Temporal Interactions in Spiking Transformers

Spiking Transformers have shown strong representation capability while consuming substantially less energy than their ANN counterparts. However, temporal modeling in existing methods remains limited, as it is mainly handled implicitly through spiking neuron dynamics. Existing studies on temporal interactions in spiking transformers mainly follow two directions: explicit spatiotemporal modeling and lightweight temporal enhancement. In the first direction, STSA [48] jointly models temporal and spatial interactions, improving performance on neuromorphic datasets at the cost of increased computational overhead. STAtten [18] further improves efficiency by adopting a block-wise strategy for temporal and spatial feature modeling while preserving the same computational complexity as spatial-only approaches. In the second direction, TIM [34], built on Spikformer [63], incorporates historical temporal information into the query branch with only a few additional parameters, thereby improving temporal modeling. SNN-ViT [46] introduces a dedicated Saccadic Spike Self-Attention (SSSA) mechanism to enhance whole-scene understanding through temporal interactions.

3. Preliminary

3.1. Leaky Integrate-and-Fire Neuron

Spiking neurons serve as the fundamental computational units of SNNs, with the leaky integrate-and-fire (LIF) model being the most widely adopted due to its balance between biological plausibility and computational efficiency. The LIF neuron integrates incoming input currents over time while allowing a gradual decay of the membrane potential through a leakage mechanism. When the accumulated po-

tential surpasses a predefined threshold, the neuron generates a spike and resets its state. This process can be formally described by the following equations:

$$\mathbf{U}_t = \lambda \mathbf{H}_{t-1} + \mathbf{I}_t \tag{1}$$

$$\mathbf{S}_t = \Theta\left(\mathbf{U}_t - \mathbf{V}_{th}\right) \tag{2}$$

$$\mathbf{H}_t = \mathbf{U}_t(1 - \mathbf{S}_t) + V_{\text{reset}}\mathbf{S}_t \tag{3}$$

Here, t denotes the discrete time step. $\mathbf{I}$ represents the input current, and $\mathbf{U}$ denotes the membrane potential after integration but prior to spike generation. $\mathbf{H}$ indicates the membrane potential after firing, which decays exponentially under the influence of the leaky factor λ. $\Theta(\cdot)$ is the Heaviside step function responsible for spike activation. When $\mathbf{U}$ exceeds the firing threshold V_{th}, the neuron emits a spike, and its membrane potential is reset to V_{reset}; otherwise, no spike is generated. For brevity, we denote the spiking neuron layer as $\mathcal{SN}(\cdot)$ throughout the following sections.

3.2. Spiking Self-Attention

Spikformer [63] pioneers the design of Spiking Self-Attention (SSA) as the first attention mechanism explicitly tailored for SNNs. Given binary queries ($\mathbf{Q}$), keys ($\mathbf{K}$), and values ($\mathbf{V}$) derived from the input tensor $\mathbf{X} \in \mathbb{R}^{T\times N\times D}$, where T, N and D denote the numbers of time steps, tokens, and feature dimensions, respectively, the SSA mechanism can be formulated as:

$$\begin{aligned}
\mathbf{Q} &= \mathcal{SN}(\mathrm{BN}(\mathbf{W_Q X})),\\
\mathbf{K} &= \mathcal{SN}(\mathrm{BN}(\mathbf{W_K X})),\\
\mathbf{V} &= \mathcal{SN}(\mathrm{BN}(\mathbf{W_V X})),\\
\mathbf{Z} &= \mathcal{SN}(\alpha \cdot \mathbf{Q}\mathbf{K}^\top \mathbf{V}),
\end{aligned} \tag{4}$$

Here, $\mathrm{BN}(\cdot)$ denotes batch normalization, $\mathbf{W_Q}$, $\mathbf{W_K}$ and $\mathbf{W_V}$ are learnable linear projection matrices for $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$, respectively, and α is a scaling factor. $\mathcal{SN}(\cdot)$ denotes the spiking neuron layer, and $\mathbf{Z}$ represents the final output of SSA. Notably, temporal dependencies in SSA are handled implicitly through the intrinsic dynamics of spiking neurons, without explicit temporal interaction modeling, which limits its temporal modeling capability.

4. Methodology

4.1. Problem Analysis

Despite significant advances in Spiking Transformers, existing spiking self-attention mechanisms lack explicit modeling of temporal dependencies. To better analyze the temporal interaction patterns within attention mechanisms, we propose a new metric, the *Temporal Interaction Coefficient* (TIC), drawing inspiration from the analytical framework in [61]. At timestep q, the TIC is defined as:

$$\mathrm{TIC}_q = \mathcal{H}(\widetilde{\mathbf{R}}_q), \tag{5}$$

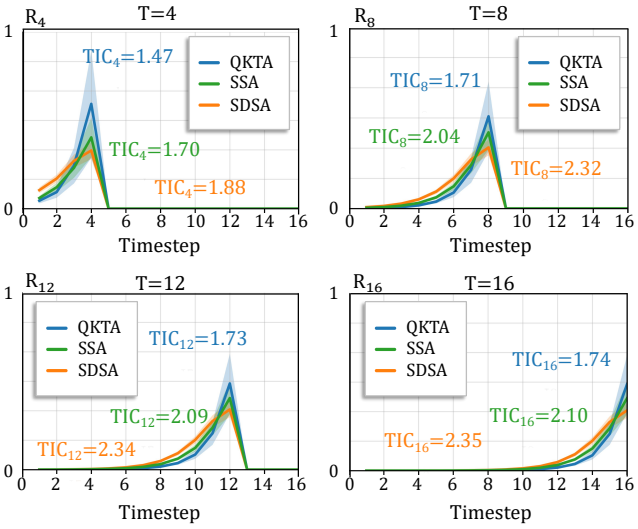


Figure 1. Visualization of reliance vectors for SSA, QKTA, and SDSA attention mechanisms, based on the CIFAR10-DVS dataset.

where $\mathcal{H}(\cdot)$ denotes the information entropy [29]. The reliance distribution $\widetilde{\mathbf{R}}_q$ is obtained by applying L_1 normalization to the reliance vector $\mathbf{R}_q = [R_{q\to 1}, \ldots, R_{q\to q}]$. Each temporal reliance coefficient $R_{q\to p}$ measures the dependency of the attention output $\mathbf{Z}$ at timestep q with respect to the input representation $\mathbf{X}$ from the current or preceding timestep p, which is computed as:

$$R_{q\to p} = \left\| \frac{\partial \mathbf{Z}_q}{\partial \mathbf{X}_p} \right\|_1, \quad \text{for } p \le q. \tag{6}$$

A higher TIC_q indicates broader temporal reliance and richer temporal interactions, whereas a lower TIC_q reflects more concentrated dependence on only a few timesteps (see Supplementary Material for details).

Building on the above definitions, we visualize the temporal reliance vectors $[\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_{16}]$ for SSA [63], QKTA [62], and SDSA [53] on the CIFAR10-DVS dataset. Representative reliance vectors $\mathbf{R}_4$, $\mathbf{R}_8$, $\mathbf{R}_{12}$, and $\mathbf{R}_{16}$ are illustrated in Fig. 1. The complete visualizations are provided in the Supplementary Material. Empirically, we observe that, for all baseline attention mechanisms, the reliance distributions are sharply peaked at the current timestep, while contributions from distant timesteps are almost absent. This narrow temporal interaction range leads to consistently low TIC values across timesteps, indicating limited temporal integration capability and potentially hindering performance.

4.2. Multi-Delay Mixer

In biological neural systems, axonal structures naturally enable temporal processing by introducing diverse transmission delays [43], as illustrated in Fig. 2(a). Such delay-based modulation enables neurons to integrate information

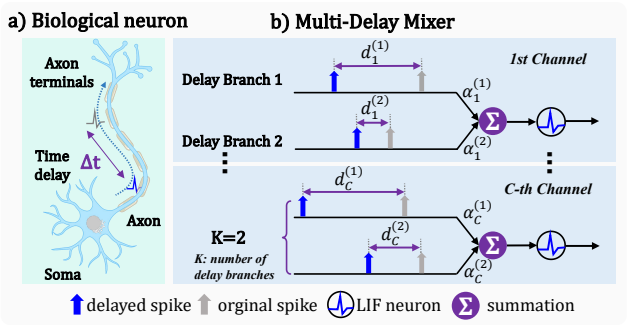


Figure 2. The proposed MD-Mixer inspired by axonal transmission delays. a) Biological inspiration. b) MD-Mixer with multiple delay branches and channel-wise aggregation, where $K = 2$ is shown for illustration.

across different temporal scales, facilitating richer temporal representations. Inspired by this mechanism, we propose the *Multi-Delay Mixer* (MD-Mixer) to address the insufficient temporal interaction modeling revealed in the previous analysis by integrating delayed information from different timesteps within spiking self-attention.

Specifically, MD-Mixer performs *channel-wise temporal aggregation* by introducing multiple learnable delay branches with adaptive weighting mechanisms, as illustrated in Fig. 2(b). For each channel $i \in \{1, \ldots, C\}$, where C denotes the number of channels, a set of K learnable delays $\{d_i^{(k)}\}_{k=1}^{K}$ is employed to establish diverse temporal patterns across different timesteps. The delayed features from multiple branches are then weighted by learnable coefficients $\alpha_i^{(k)}$ and aggregated, enabling adaptive modulation of temporal information. Formally, the channel-wise temporal aggregation can be expressed as:

$$\widetilde{\mathbf{X}}_{t,i} = \sum_{k=1}^{K} \alpha_i^{(k)} \, \mathbf{X}_{t-d_i^{(k)},i}, \tag{7}$$

where $\mathbf{X}_{t,i}$ denotes the input tensor at time step t for channel i, $\widetilde{\mathbf{X}}_{t,i}$ represents the aggregated temporal feature, and both $d_i^{(k)}$ and $\alpha_i^{(k)}$ are jointly optimized during training to adaptively capture multi-scale temporal dependencies. The optimization strategy for delays is detailed in Section 4.4. Unlike approaches that rely on predefined temporal windows or recurrent connections, MD-Mixer employs channel-specific learnable delays to construct enriched temporal representations with multi-scale dependencies, achieving data-driven and parameter-efficient temporal modeling while preserving the event-driven nature of SNNs. This property makes it naturally compatible with existing spiking self-attention mechanisms to address their temporal modeling limitations.

4.3. Multi-Delay Self-Attention Framework

By integrating the proposed MD-Mixer into existing spiking transformer architectures, we develop the unified *Multi-Delay Self-Attention Framework*, as illustrated in Fig. 3(a). This framework enables temporal mixing through delay-based temporal interactions, thereby achieving space–time mixing by applying spatial attention to temporally integrated features.

Temporal Mixing. The MD-Mixer performs multi-scale temporal integration within each channel, thereby generating delay-enhanced $\mathbf{K}$ and $\mathbf{V}$ that encode rich temporal dynamics, while $\mathbf{Q}$ maintain the original temporal alignment. Formally, given an input tensor $\mathbf{X} \in \mathbb{R}^{T\times N\times C}$, the query, key, and value projections are obtained as:

$$\mathbf{Q} = \mathcal{SN}(\mathrm{BN}(\mathrm{Linear}(\mathbf{X}))), \tag{8}$$

$$\mathbf{K} = \mathcal{SN}(\mathrm{BN}(\text{MD-Mixer}(\mathbf{X}))), \tag{9}$$

$$\mathbf{V} = \mathcal{SN}(\mathrm{BN}(\text{MD-Mixer}(\mathbf{X}))). \tag{10}$$

Here, $\mathrm{BN}(\cdot)$ denotes batch normalization, and $\mathrm{Linear}(\cdot)$ represents a fully connected layer. After temporal mixing, the representations carry temporally enriched context, facilitating space–time modeling.

Space-Time Mixing. The obtained $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are subsequently fed into the spiking self-attention module, where the attention computation extends from purely spatial correlations to spatiotemporal interactions. To illustrate this, Fig. 3(b) visualizes the vanilla $\mathbf{K}/\mathbf{V}$ features used in the baseline attention, which are computed independently at each timestep and therefore capture only spatial correlations. In contrast, the delayed $\mathbf{K}/\mathbf{V}$ features in Fig. 3(c) aggregate contextual information from current and preceding timesteps, achieving space–time mixing in the attention computation. Formally, the attention output is given by

$$\mathbf{Z} = \mathrm{Atten}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \tag{11}$$

where $\mathbf{Z}$ denotes the attention output, and $\mathrm{Atten}(\cdot)$ denotes a generic attention function. With the delay-based temporal interactions provided by the MD-Mixer, the spiking self-attention evolves from spatial-only modeling to *space–time mixing*, capturing both spatial and temporal dependencies in spike-driven computation.

4.4. Optimization of Delays

To facilitate the learning of temporal delays, we introduce a triangular delay distribution $\Phi = \{\phi(d; d^*)\}_{d=0}^{T-1}$ with each element defined as

$$\phi(d; d^*) = \mathcal{N}\left(\sigma\left(1 - \frac{|d - d^*|}{\tau}\right)\right). \tag{12}$$

Here, $\phi(d; d^*)$ represents the probability weight assigned to delay d, d^* controls the center of the distribution, τ

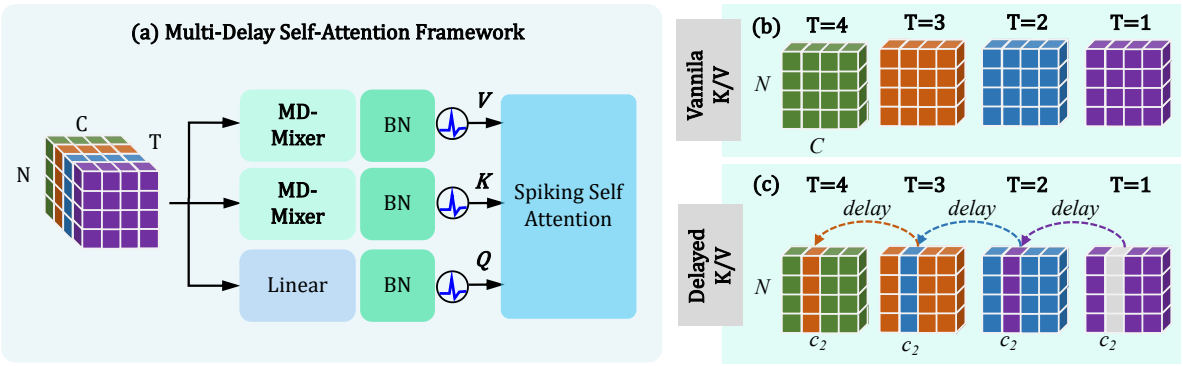


Figure 3. Illustration of the multi-delay self-attention framework. (a) The framework employs two MD-Mixer modules to generate keys (K) and values (V), and a linear projection for queries (Q), followed by batch normalization (BN) and spiking self-attention. (b) Keys and values of vanilla self-attention across timesteps (T=1 to T=4). (c) The delayed keys and values capture multi-scale temporal dependencies by aggregating historical information at learnable temporal delays.

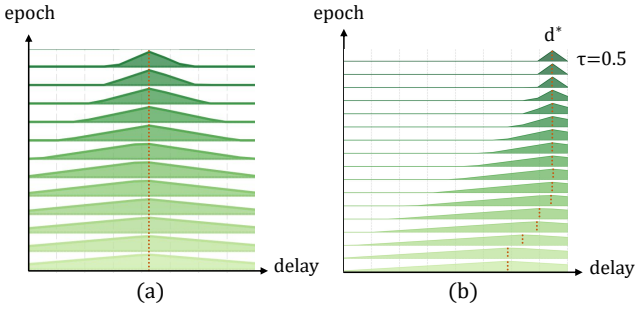


Figure 4. Visualization of delay optimization strategy. (a) Progressive sharpening of the distribution as temperature τ anneals across training epochs. (b) Joint optimization of distribution sharpness and the learnable delay center d^*.

determines its sharpness, $\mathcal{N}(\cdot)$ denotes normalization over all candidate delays, and $\sigma(\cdot) = \max(0, \cdot)$ represents the ReLU activation function. As τ decreases during training, the distribution gradually narrows and becomes increasingly concentrated on a discrete delay.

With the delay distribution properly established, we now incorporate it into the spiking neuron dynamics in MD-Mixer. Specifically, for the j-th output channel, the membrane potential is updated as:

$$\mathbf{U}_{t,j} = \lambda \mathbf{H}_{t-1,j} + \mathbf{I}_{t,j}, \tag{13}$$

$$\mathbf{I}_{t,j} = \sum_{k=1}^{K} \alpha_j^{(k)} \sum_{d=0}^{T-1} \phi(d; d_j^{(k),*}) \cdot \mathbf{X}_{t-d,j}. \tag{14}$$

Here, the delay weight $\phi(d; d_j^{(k),*})$ modulates the contribution of historical features with lag d, determining the input current $\mathbf{I}_{t,j}$ at the present timestep t.

During training, the temperature parameter τ undergoes progressive annealing following a squared cosine schedule, which facilitates a soft-to-hard transition in delay learning. As illustrated in Fig. 4(a), this annealing mechanism induces a gradual sharpening of the delay distribution Φ: initially, when τ is large, the distribution maintains a broad profile that enables exploration across multiple temporal scales; as τ decreases, the distribution progressively concentrates to a sharp peak. At convergence, Φ approximates a one-hot distribution centered at the optimal delay, enabling efficient discrete delay learning through continuous gradient-based optimization, as illustrated in Fig. 4(b). Detailed gradient analysis and the temperature annealing schedule are provided in the Supplementary Material.

4.5. Computational Efficiency

Analysis of the MD-Mixer's computational complexity provides insights into its efficiency characteristics. As shown in Eq. 7, the module reduces both computational complexity from $\mathcal{O}(TD^2)$ to $\mathcal{O}(KTD)$ and parameter count from D^2 to KD, yielding a reduction factor of approximately D/K in both cases. Given the architectural design where $K \ll D$ (e.g., $K = 5$ while $D = 256$), the MD-Mixer theoretically achieves efficiency gains over standard linear layers used for Key and Value computation in attention mechanisms, demonstrating that temporal modeling can be incorporated without substantial computational overhead.

5. Experiment

We validate MD-Mixer through extensive experiments on static, neuromorphic, and long-term datasets across various Spiking Transformer architectures. This section is organized as follows: First, we describe the experimental settings. Second, we evaluate MD-Mixer's performance by integrating it into existing Spiking Transformer architectures and comparing it with state-of-the-art methods. Finally, we conduct ablation studies to verify the effectiveness of the delay learning mechanism and analyze hyperparameters.

Table 1. Performance comparison on ImageNet dataset across various spiking Transformer architectures. Notes: L-D represents L encoder blocks with D hidden dimensions; * denotes results with 288 $\times$ 288 input resolution during inference.

Method	Architecture	Params(M)	T	Accuracy(%)
Spikformer [63]	Spikformer-8-768	66.34	4	74.81
SpikingReformer [38]	SpikingReformer-6-1024	66.38	4	78.77 / 79.40*
QKFormer [62]	QKFormer-6-1024	64.96	4	84.22
Spike-driven Transformer V1 [53]	Spike-driven Transformer V1-8-512	29.68	4	74.57
	Spike-driven Transformer V1-8-768	66.34	4	76.32 / 77.07*
	Spike-driven Transformer V1-10-512	36.01	4	74.66
STAtten [18] + [53]	Spike-driven Transformer V1-8-512	29.68	4	76.18 / 76.56*
	Spike-driven Transformer V1-8-768	66.34	4	78.11 / 78.39*
MD-Mixer + [53]	Spike-driven Transformer V1-8-512	29.68	4	76.27 / 76.87*
	Spike-driven Transformer V1-8-768	66.34	4	**78.23 / 78.53***
Spike-driven Transformer V2 [56]	Spike-driven Transformer V2-8-512	55.4	4	79.49 / 79.98*
STAtten [18] + [56]	Spike-driven Transformer V2-8-512	55.4	4	79.85 / 80.67*
MD-Mixer + [56]	Spike-driven Transformer V2-8-512	55.4	4	**80.02 / 80.73***

5.1. Experimental Settings

For performance comparison, we integrate MD-Mixer into four representative Spiking Transformer architectures: Spikformer [63], QKFormer [62], Spike-driven Transformer V1 [53] and Spike-driven Transformer V2 [56]. In ablation studies, all experiments are conducted on the Spike-driven Transformer V1 baseline. Detailed model configurations, including optimizer settings, learning rate, and architecture hyper-parameters, are provided in the Supplementary Material.

5.2. Performance Comparison

5.2.1. Static Dataset

We evaluate MD-Mixer on ImageNet [6], comprising approximately 1.3 million training images and 50,000 validation images across 1,000 classes, using synaptic-wise delays. Table 1 presents performance comparisons across various Spiking Transformer architectures. When integrated into Spike-driven Transformer V1, MD-Mixer consistently delivers substantial performance gains across different model scales, significantly outperforming the baseline while maintaining competitive performance with STAtten [18]. Specifically, under the 8–768 configuration at 224 $\times$ 224 resolution, MD-Mixer achieves 78.23% top-1 accuracy, corresponding to a notable 1.91% improvement over the baseline and remaining on par with STAtten (a 0.12% gain). When deployed with Spike-driven Transformer V2 in the 8-512 configuration, MD-Mixer attains 80.02% top-1 accuracy at 224 $\times$ 224 resolution, achieving a 0.53% improvement over the baseline. These results convincingly demonstrate the efficacy of our temporal modeling approach on static datasets.

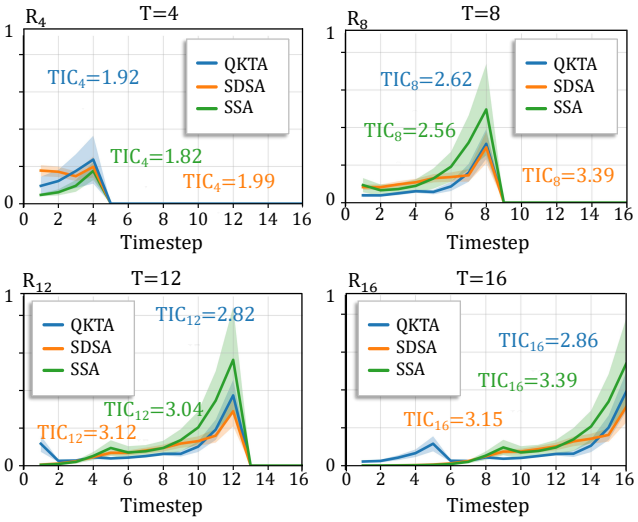


Figure 5. Visualization of reliance vectors for SSA, QKTA, and SDSA attention mechanisms integrated with MD-Mixer, based on the CIFAR10-DVS dataset.

5.2.2. Neuromorphic Dataset

CIFAR10-DVS and N-Caltech101 CIFAR10-DVS contains 10,000 DVS-recorded samples at 128 $\times$ 128 resolution with 9,000 training and 1,000 test samples, while N-Caltech101 comprises 8,831 samples across 101 categories at 180 $\times$ 240 resolution. Following [18, 23], both datasets are resized to 64 $\times$ 64. We adopt augmentation strategies from [53] for CIFAR10-DVS and [23] for N-Caltech101. Table 2 reports the results on CIFAR10-DVS and N-Caltech101 datasets. Integrating the proposed MD-Mixer into various architectures consistently

Table 2. Performance comparison on CIFAR10-DVS and N-Caltech101 datasets. SDT-V1 denotes Spike-driven Transformer V1.

Method	Architecture	Timestep	CIFAR10-DVS (%)	N-Caltech101 (%)
Spikformer [63]	Spikformer-2-256	16	80.90	80.23
TIM + [63]	Spikformer-2-256	16	82.00	82.30
STAtten + [63]	Spikformer-2-256	16	82.40	83.12
MD-Mixer + [63]	Spikformer-2-256	16	**83.37**	**84.57**
QKFormer [62]	QKFormer-2-256	16	84.10	79.69
STAtten + [62]	QKFormer-2-256	16	84.48	81.22
MD-Mixer + [62]	QKFormer-2-256	16	**85.20**	**82.06**
SDT-V1 [53]	SDT-V1-2-256	16	80.00	81.80
STAtten + [53]	SDT-V1-2-256	16	81.10	83.15
MD-Mixer + [53]	SDT-V1-2-256	16	**82.40**	**84.59**

improves classification accuracy over baselines. Specifically, our method achieves 83.37% and 84.57% on Spikformer, 85.20% and 82.06% on QKFormer, and 82.40% and 84.59% on Spike-driven Transformer V1 for CIFAR10-DVS and N-Caltech101, respectively. Notably, integrating MD-Mixer into QKFormer achieves a 1.10% improvement on CIFAR10-DVS, while its integration with Spike-driven Transformer V1 brings a 2.79% gain on N-Caltech101, achieving new state-of-the-art results among Spiking Transformer on both datasets. As shown in Fig. 5, integrating MD-Mixer produces broader reliance vectors compared to the sharply peaked patterns of baseline mechanisms (Fig. 1). MD-Mixer substantially improves TIC values across all timesteps, with TIC_{16} increasing from baseline values of 1.74, 2.10, and 2.35 to 2.86, 3.39, and 3.15 for QKTA, SSA, and SDSA, respectively. This demonstrates effective integration of historical information and the effectiveness of our multi-delay design on temporal interaction.

UCF101-DVS and HMDB51-DVS UCF101-DVS and HMDB51-DVS are respectively captured from playbacks of the UCF101 [42] and HMDB [16] datasets by DVS camera. UCF101-DVS includes 101 categories of human actions, and HMDB51-DVS comprises 51 categories. The comparison results on UCF101-DVS and HMDB51-DVS are summarized in Table 3. When combined with the Spike-driven Transformer V2, our method achieves state-of-the-art performance, obtaining 66.3% and 62.8% accuracy on UCF101-DVS and HMDB51-DVS, respectively. These results outperform the previous best method TIM [34] by 2.5% and 4.2%, respectively.

5.3. Sequential CIFAR10/100 Classification

To further validate the effectiveness of the proposed MD-Mixer in long-term temporal pattern modeling, we conduct experiments on the Sequential CIFAR-10/100 (s-CIFAR10/100) datasets. Each sequence sample comprises 32 timesteps, with each step corresponding to a column of the original CIFAR-10 or CIFAR-100 image. The results

Table 3. Performance comparison on UCF101-DVS and HMDB51-DVS datasets. SDT-V2 denotes Spike-driven Transformer V2.

Dataset	Model	Acc (%)
UCF101-DVS	P3D-63 [31]	53.4
	C3D [44]	47.2
	Res-SNN18 + RM [54]	63.5
	I3D [3]	63.5
	TIM [34]	63.8
	MD-Mixer + SDT-V2	**66.3**
HMDB51-DVS	C3D [44]	41.7
	P3D-63 [31]	40.4
	Res-SNN18 + RM [54]	44.7
	Spikepoint [32]	55.6
	TIM [34]	58.6
	MD-Mixer + SDT-V2	**62.8**

Table 4. Performance comparison on the s-CIFAR10 and s-CIFAR100 datasets. All values are percentages. SDT-V1 denotes Spike-driven Transformer V1.

Method	s-CIFAR10	s-CIFAR100
Spikformer [63]	76.94	55.65
TIM + [63]	77.02	56.35
STAtten + [63]	77.96	56.60
MD-Mixer + [63]	**78.74**	**59.41**
QKFormer [62]	79.48	55.99
STAtten + [62]	79.54	56.23
MD-Mixer + [62]	**85.09**	**64.33**
SDT-V1 [53]	83.65	61.54
STAtten + [53]	83.90	61.07
MD-Mixer + [53]	**86.65**	**62.44**

on the s-CIFAR10 and s-CIFAR100 datasets are summarized in Table 4. Across three representative architectures, integrating the proposed MD-Mixer consistently improves performance. Specifically, when applied to Spikformer,

our method achieves 78.74% and 59.41% accuracy on s-CIFAR10 and s-CIFAR100, surpassing STAtten [17] by 0.78% and 2.81%, respectively. When integrated with QK-Former, the performance further increases to 85.09% and 64.33%, with the latter establishing a new state-of-the-art result on s-CIFAR100. Finally, combined with the Spike-driven Transformer V1, our method reaches 86.65% on s-CIFAR10 and 62.44% on s-CIFAR100, achieving the best performance reported so far on s-CIFAR10. These results demonstrate the effectiveness of MD-Mixer in modeling long-term temporal interactions and enhancing overall performance. Notably, compared with neuromorphic datasets, our method shows more pronounced improvements on the Sequential benchmarks. This can be attributed to the fact that the s-CIFAR10/100 datasets exhibit richer and more structured temporal dynamics than the DVS-captured neuromorphic datasets.

5.4. Ablation Study

5.4.1. Effect of Learnable Delays

To assess the impact of learnable delays, we compare two configurations: random delays with fixed, randomly initialized values, and MD-Mixer with delays optimized during training. Aggregation coefficients remain learnable in both settings. Table 5 summarizes the results on s-CIFAR10 and CIFAR10-DVS datasets. Random delays bring modest performance gains of 0.74% and 0.52% over the baseline on s-CIFAR10 and CIFAR10-DVS, respectively. In contrast, MD-Mixer achieves substantially larger improvements of 3.00% and 2.44%, respectively. These results demonstrate that while random delays provide slight improvements through temporal diversity, learnable delays in MD-Mixer fundamentally enable adaptive capture and optimization of multi-scale temporal patterns, thereby achieving significantly superior performance. Visualization of the learned delays is provided in the Supplementary Material.

5.4.2. Effect of the Number of Delay Branches

We evaluate the impact of the number of delay branches on model performance across s-CIFAR10 and CIFAR10-DVS datasets. As illustrated in Fig. 6(a), the number of delay branches has a clear impact on model performance across both datasets. On s-CIFAR10, the accuracy increases from 84.3% with a single branch to a peak of 86.7% with three branches, then gradually declines to 84.2% with six branches. Similarly, on CIFAR10-DVS, performance improves from 81.1% with one branch to 82.4% with four branches, before dropping to 80.9% with six branches. These results indicate that an optimal number of delay branches most effectively captures multi-scale temporal patterns, while excessive branches impair performance due to overly complex temporal modeling.

Table 5. Ablation study on learnable delays. Random Delay uses fixed random delays, while MD-Mixer learns optimal delays.

Dataset	Model	Accuracy(%)
s-CIFAR10	Baseline	83.65
	+ Random Delay	84.39
	+ MD-Mixer	**86.65**
CIFAR10-DVS	Baseline	80.00
	+ Random Delay	80.52
	+ MD-Mixer	**82.44**

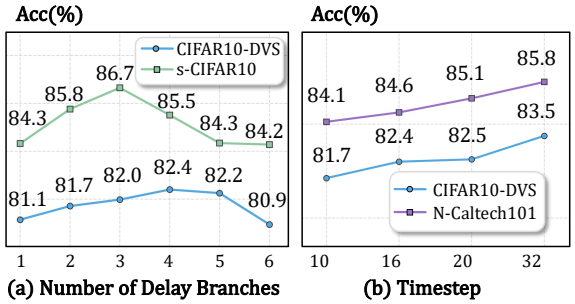


Figure 6. Impact of hyperparameters. (a) Number of delay branches on CIFAR10-DVS and s-CIFAR10. (b) Number of timesteps on CIFAR10-DVS and N-Caltech101.

5.4.3. Effect of Timestep

We evaluate the effect of timestep on the CIFAR10-DVS and N-Caltech101 datasets. The experimental results in Fig. 6(b) show a clear trend in the effect of timestep settings on model performance. As the number of timesteps increases, accuracy improves for both CIFAR10-DVS and N-Caltech101 datasets. Specifically, on CIFAR10-DVS, accuracy rises from 81.7% at 10 timesteps to 83.5% at 32 timesteps, while on N-Caltech101, accuracy increases from 84.1% at 10 timesteps to 85.8% at 32 timesteps.

6. Conclusion

In this paper, we introduce a novel metric TIC to evaluate the temporal interaction capability of existing spiking self-attention mechanisms and visualize their temporal dependency patterns. Our evaluation reveals limited temporal interactions that hinder their performance. Inspired by biological time delay, we propose MD-Mixer with learnable multi-scale temporal delays that seamlessly integrate into existing spiking transformers. By coupling multi-scale temporal patterns with self-attention, MD-Mixer achieves comprehensive space-time mixing and significantly enhances spatiotemporal feature learning. Extensive experiments demonstrate substantial improvements across various architectures on both static and neuromorphic datasets. We hope this work inspires future research on biologically-plausible temporal modeling in neuromorphic computing.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grants 62576080 and 62220106008), the Sichuan Science and Technology Program (Grant 2024NSFTD0034), and the Research Grants Council of the Hong Kong SAR (PolyU25216423 and PolyU15217424).

References

[1] Filipp Akopyan, Jun Sawada, Andrew Cassidy, Rodrigo Alvarez-Icaza, John Arthur, Paul Merolla, Nabil Imam, Yutaka Nakamura, Pallab Datta, Gi-Joon Nam, et al. Truenorth: Design and tool flow of a 65 mw 1 million neuron programmable neurosynaptic chip. *IEEE transactions on computer-aided design of integrated circuits and systems*, 34 (10):1537–1557, 2015. 1

[2] Yongqiang Cao, Yang Chen, and Deepak Khosla. Spiking deep convolutional neural networks for energy-efficient object recognition. *International Journal of Computer Vision*, 113(1):54–66, 2015. 2

[3] João Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4724–4733, 2017. 7

[4] Stefano Caviglia, Maurizio Valle, and Chiara Bartolozzi. Asynchronous, event-driven readout of posfet devices for tactile sensing. In *2014 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 2648–2651. IEEE, 2014. 1

[5] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *Ieee Micro*, 38(1):82–99, 2018. 1

[6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 2, 6

[7] Wei Fang, Zhaofei Yu, Yanqi Chen, Tiejun Huang, Timothée Masquelier, and Yonghong Tian. Deep residual learning in spiking neural networks. *Advances in Neural Information Processing Systems*, 34:21056–21069, 2021. 2

[8] Wei Fang, Zhaofei Yu, Yanqi Chen, Timothée Masquelier, Tiejun Huang, and Yonghong Tian. Incorporating learnable membrane time constant to enhance learning of spiking neural networks. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2661–2671, 2021. 1, 2

[9] Wulfram Gerstner and Werner M Kistler. *Spiking neuron models: Single neurons, populations, plasticity*. Cambridge university press, 2002. 1

[10] Yufei Guo, Yuanpei Chen, Xiaode Liu, Weihang Peng, Yuhan Zhang, Xuhui Huang, and Zhe Ma. Ternary spike: Learning ternary spikes for spiking neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, pages 12244–12252, 2024. 1

[11] Yufei Guo, Xiaode Liu, Yuanpei Chen, Weihang Peng, Yuhan Zhang, and Zhe Ma. Spiking transformer: Introducing accurate addition-only spiking self-attention for transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24398–24408, 2025. 1

[12] Bing Han, Gopalakrishnan Srinivasan, and Kaushik Roy. Rmp-snn: Residual membrane potential neuron for enabling deeper high-accuracy and low-latency spiking neural network. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13558–13567, 2020. 2

[13] Yifan Hu, Lei Deng, Yujie Wu, Man Yao, and Guoqi Li. Advancing spiking neural networks toward deep residual learning. *IEEE transactions on neural networks and learning systems*, 36(2):2353–2367, 2024. 2

[14] Eugene M Izhikevich. Simple model of spiking neurons. *IEEE Transactions on neural networks*, 14(6):1569–1572, 2003. 1

[15] Amirhossein Javanshir, Thanh Thi Nguyen, MA Parvez Mahmud, and Abbas Z Kouzani. Advancements in algorithms and neuromorphic hardware for spiking neural networks. *Neural Computation*, 34(6):1289–1328, 2022. 2

[16] Hildegard Kuehne, Hueihan Jhuang, Estíbaliz Garrote, Tomaso Poggio, and Thomas Serre. Hmdb: a large video database for human motion recognition. In *2011 International conference on computer vision*, pages 2556–2563. IEEE, 2011. 7

[17] Donghyun Lee, Yuhang Li, Youngeun Kim, Shiting Xiao, and Priyadarshini Panda. Spiking transformer with spatial-temporal attention. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13948–13958, 2025. 8

[18] Donghyun Lee, Yuhang Li, Youngeun Kim, Shiting Xiao, and Priyadarshini Panda. Spiking transformer with spatial-temporal attention. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 13948–13958, 2025. 1, 2, 6

[19] Zhenxin Lei, Man Yao, Jiakui Hu, Xinhao Luo, Yanye Lu, Bo Xu, and Guoqi Li. Spike2former: Efficient spiking transformer for high-performance image segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1364–1372, 2025. 1

[20] Hongmin Li, Hanchao Liu, Xiangyang Ji, Guoqi Li, and Luping Shi. Cifar10-dvs: an event-stream dataset for object classification. *Frontiers in neuroscience*, 11:244131, 2017. 2

[21] Jindong Li, Guobin Shen, Dongcheng Zhao, Qian Zhang, and Yi Zeng. Firefly: A high-throughput hardware accelerator for spiking neural networks with efficient dsp and memory optimization. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 31(8):1178–1191, 2023. 1

[22] Jindong Li, Guobin Shen, Dongcheng Zhao, Qian Zhang, and Yi Zeng. Firefly v2: Advancing hardware support for high-performance spiking neural network with a spatiotemporal fpga accelerator. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 43(9): 2647–2660, 2024. 1

[23] Yuhang Li, Youngeun Kim, Hyoungseob Park, Tamar Geller, and Priyadarshini Panda. Neuromorphic data augmentation for training spiking neural networks. In *European Conference on Computer Vision*, pages 631–649. Springer, 2022. 6

[24] Hanwen Liu, Yi Chen, Zihang Zeng, Malu Zhang, and Hong Qu. A low power and low latency fpga-based spiking neural network accelerator. In *2023 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2023. 1

[25] Hanwen Liu, Kexin Shi, Wenyu Chen, Malu Zhang, and Yang Yang. Jqa: Joint design of hardware-friendly quantization and efficient accelerator for spiking neural networks. In *2025 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND)*, pages 162–167. IEEE, 2025. 1

[26] Hanwen Liu, Kexin Shi, Jieyuan Zhang, Yimeng Shan, Jibin Wu, Wenyu Chen, and Malu Zhang. Hardf-snn: Hardware-friendly quantization for spiking neural networks with efficient integer-arithmetic-only inference. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(3):1937–1945, 2026. 2

[27] Wolfgang Maass. Networks of spiking neurons: the third generation of neural network models. *Neural networks*, 10 (9):1659–1671, 1997. 1

[28] Garrick Orchard, E Paxon Frady, Daniel Ben Dayan Rubin, Sophia Sanborn, Sumit Bam Shrestha, Friedrich T Sommer, and Mike Davies. Efficient neuromorphic signal processing with loihi 2. In *2021 IEEE Workshop on Signal Processing Systems (SiPS)*, pages 254–259. IEEE, 2021. 1

[29] Liam Paninski. Estimation of entropy and mutual information. *Neural computation*, 15(6):1191–1253, 2003. 3

[30] Jing Pei, Lei Deng, Sen Song, Mingguo Zhao, Youhui Zhang, Shuang Wu, Guanrui Wang, Zhe Zou, Zhenzhi Wu, Wei He, et al. Towards artificial general intelligence with hybrid tianjic chip architecture. *Nature*, 572(7767):106–111, 2019. 1

[31] Zhaofan Qiu, Ting Yao, and Tao Mei. Learning spatio-temporal representation with pseudo-3d residual networks. *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 5534–5542, 2017. 7

[32] Hongwei Ren, Yue Zhou, Yulong Huang, Haotian FU, Jie Song, Bojun Cheng, et al. Spikepoint: An efficient point-based spiking neural network for event cameras action recognition. In *The Twelfth International Conference on Learning Representations*, 2024. 7

[33] Haibo Shen, Yihao Luo, Xiang Cao, Liangqi Zhang, Juyu Xiao, and Tianjiang Wang. Training robust spiking neural networks on neuromorphic data with spatiotemporal fragments. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2023. 1

[34] Sicheng Shen, Dongcheng Zhao, Guobin Shen, and Yi Zeng. Tim: An efficient temporal interaction module for spiking transformer. In *IJCAI*, 2024. 1, 2, 7

[35] Kexin Shi, Hanwen Liu, Yi Chen, and Hong Qu. Hl-esvit: High-low frequency efficient spiking vision transformer. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2024. 1

[36] Kexin Shi, Mengshu Hou, Xiaoling Luo, Dehao Zhang, Hanwen Liu, and Jingya Wang. Spike-vad: Efficient and robust spiking neural network for voice activity detection. *IEEE Transactions on Cognitive and Developmental Systems*, 2025. 1

[37] Kexin Shi, Hanwen Liu, Jibin Wu, Wenyu Chen, Malu Zhang, and Yang Yang. Seend: A spike-based end-to-end framework for energy-efficient speaker diarization. In *2025 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND)*, pages 168–173. IEEE, 2025. 2

[38] Xinyu Shi, Zecheng Hao, and Zhaofei Yu. Spikingresformer: Bridging resnet and vision transformer in spiking neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5610–5619, 2024. 1, 2, 6

[39] Zeyang Song, Qianhui Liu, Qu Yang, Yizhou Peng, and Haizhou Li. Ed-skws: Early-decision spiking neural networks for rapid, and energy-efficient keyword spotting. *arXiv preprint arXiv:2406.12726*, 2024. 2

[40] Zeyang Song, Jibin Wu, Malu Zhang, Mike Zheng Shou, and Haizhou Li. Spiking-leaf: A learnable auditory front-end for spiking neural networks. In *ICASSP 2024-2024 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 226–230. IEEE, 2024. 2

[41] Zeyang Song, Shimin Zhang, Yuhong Chou, Jibin Wu, and Haizhou Li. Iml-spikeformer: Input-aware multilevel spiking transformer for speech processing. *IEEE Transactions on Neural Networks and Learning Systems*, 2025. 1

[42] Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. Ucf101: A dataset of 101 human actions classes from videos in the wild. *arXiv preprint arXiv:1212.0402*, 2012. 7

[43] Carl R Stoelzel, Yulia Bereshpolova, Jose-Manuel Alonso, and Harvey A Swadlow. Axonal conduction delays, brain state, and corticogeniculate communication. *Journal of Neuroscience*, 37(26):6342–6358, 2017. 3

[44] Du Tran, Lubomir D. Bourdev, Rob Fergus, Lorenzo Torresani, and Manohar Paluri. Learning spatiotemporal features with 3d convolutional networks. *2015 IEEE International Conference on Computer Vision (ICCV)*, pages 4489–4497, 2014. 7

[45] Shuai Wang, Malu Zhang, Jingya Wang, Dehao Zhang, Yimeng Shan, Jieyuan Zhang, Yichen Xiao, Honglin Cao, Haonan Zhang, Zeyu Ma, et al. Bipolar self-attention for spiking transformers. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 1

[46] Shuai Wang, Malu Zhang, Dehao Zhang, Ammar Belatreche, Yichen Xiao, Yu Liang, Yimeng Shan, Qian Sun, Enqi Zhang, and Yang Yang. Spiking vision transformer with saccadic attention. In *The Thirteenth International Conference on Learning Representations*, 2025. 2

[47] Shuai Wang, Haorui Zheng, Yukun Chen, Ammar Belatreche, Guoqing Wang, Yeying Jin, Jibin Wu, Malu Zhang, Yang Yang, and Haizhou Li. Snn-ft: Temporal-coded spiking neural networks for fourier transform. *IEEE Transactions on Neural Networks and Learning Systems*, 2025. 2

[48] Yuchen Wang, Kexin Shi, Chengzhuo Lu, Yuguo Liu, Malu Zhang, and Hong Qu. Spatial-temporal self-attention for

asynchronous spiking neural networks. In *IJCAI*, pages 3085–3093, 2023. 1, 2

[49] Peixi Wu, Bosong Chai, Hebei Li, Menghua Zheng, Yansong Peng, Zeyu Wang, Xuan Nie, Yueyi Zhang, and Xiaoyan Sun. Spiking point transformer for point cloud classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 21563–21571, 2025. 1

[50] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in neuroscience*, 12:331, 2018. 2

[51] Yichen Xiao, Shuai Wang, Dehao Zhang, Wenjie Wei, Yimeng Shan, Xiaoli Liu, Yulin Jiang, and Malu Zhang. Rethinking spiking self-attention mechanism: Implementing a-xnor similarity calculation in spiking transformers. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5444–5454, 2025. 1

[52] Yinsong Yan, Qu Yang, Yujie Wu, Hanwen Liu, Malu Zhang, Haizhou Li, Kay Chen Tan, and Jibin Wu. Efficient and robust temporal processing with neural oscillations modulated spiking neural networks. *Nature communications*, 16 (1):8651, 2025. 1

[53] Man Yao, Jiakui Hu, Zhaokun Zhou, Li Yuan, Yonghong Tian, Bo Xu, and Guoqi Li. Spike-driven transformer. *Advances in neural information processing systems*, 36:64043–64058, 2023. 1, 2, 3, 6, 7

[54] Man Yao, Hengyu Zhang, Guangshe Zhao, Xiyu Zhang, Dingheng Wang, Gang Cao, and Guoqi Li. Sparser spiking activity can be better: Feature refine-and-mask spiking neural network for event-based visual recognition. *Neural Networks*, 166:410–423, 2023. 7

[55] Man Yao, Guangshe Zhao, Hengyu Zhang, Yifan Hu, Lei Deng, Yonghong Tian, Bo Xu, and Guoqi Li. Attention spiking neural networks. *IEEE transactions on pattern analysis and machine intelligence*, 45(8):9393–9410, 2023. 2

[56] Man Yao, JiaKui Hu, Tianxiang Hu, Yifan Xu, Zhaokun Zhou, Yonghong Tian, Bo Xu, and Guoqi Li. Spike-driven transformer v2: Meta spiking neural network architecture inspiring the design of next-generation neuromorphic chips. *arXiv preprint arXiv:2404.03663*, 2024. 1, 6

[57] Man Yao, Xuerui Qiu, Tianxiang Hu, Jiakui Hu, Yuhong Chou, Keyu Tian, Jianxing Liao, Luziwei Leng, Bo Xu, and Guoqi Li. Scaling spike-driven transformer with efficient spike firing approximation training. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 2

[58] Qiang Yu, Huajin Tang, Kay Chen Tan, and Haoyong Yu. A brain-inspired spiking neural network model with temporal encoding and learning. *Neurocomputing*, 138:3–13, 2014. 1

[59] Dehao Zhang, Shuai Wang, Ammar Belatreche, Wenjie Wei, Yichen Xiao, Haorui Zheng, Zijian Zhou, Malu Zhang, and Yang Yang. Spike-based neuromorphic model for sound source localization. *Advances in Neural Information Processing Systems*, 37:113911–113936, 2024. 2

[60] Jiqing Zhang, Bo Dong, Haiwei Zhang, Jianchuan Ding, Felix Heide, Baocai Yin, and Xin Yang. Spiking transformers for event-based single object tracking. In *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*, pages 8801–8810, 2022. 1

[61] Jieyuan Zhang, Xiaolong Zhou, Shuai Wang, Wenjie Wei, Hanwen Liu, Qian Sun, Malu Zhang, Yang Yang, and Haizhou Li. Unveiling the spatial-temporal effective receptive fields of spiking neural networks. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 3

[62] Chenlin Zhou, Han Zhang, Zhaokun Zhou, Liutao Yu, Liwei Huang, Xiaopeng Fan, Li Yuan, Zhengyu Ma, Huihui Zhou, and Yonghong Tian. Qkformer: Hierarchical spiking transformer using qk attention. *Advances in Neural Information Processing Systems*, 37:13074–13098, 2024. 1, 3, 6, 7

[63] Zhaokun Zhou, Yuesheng Zhu, Chao He, Yaowei Wang, Shuicheng YAN, Yonghong Tian, and Li Yuan. Spikformer: When spiking neural network meets transformer. In *The Eleventh International Conference on Learning Representations*, 2023. 1, 2, 3, 6, 7

[64] Rui-Jie Zhu, Malu Zhang, Qihang Zhao, Haoyu Deng, Yule Duan, and Liang-Jian Deng. Tcja-snn: Temporal-channel joint attention for spiking neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 36(3): 5112–5125, 2024. 2